Unsupervised Deep Hyperspectral Video Target Tracking and High Spectral-Spatial-Temporal Resolution (H^3) Benchmark Dataset

Zhenqi Liu, Yanfei Zhong, *Senior Member, IEEE*, Xinyu Wang, Meng Shu, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Target tracking has received increased attention in the past few decades. However, most of the target tracking algorithms are based on RGB video data, and few are based on hyperspectral video data. With the development of the new "snapshot" hyperspectral sensors, hyperspectral videos can now be easily obtained. However, hyperspectral video target tracking datasets are still rare. In this article, a high spectral-spatial-temporal resolution hyperspectral video target tracking algorithm framework (H^3Net) based on deep learning is proposed. The proposed framework consists of two main parts: 1) an unsupervised deep learning-based target tracking training framework for hyperspectral video; and 2) a dual-branch network structure based on a Siamese network. Using the dual-branch network, the H^3Net framework can utilize both the spatial and spectral information. The combination of deep learning and a discriminative correlation filter (DCF) makes the features extracted by deep learning more suitable for the DCF. Compared with hyperspectral images, hyperspectral video data require more manpower to annotate, so we propose an unsupervised approach to train H^3Net, without any annotation. To solve the problem of the lack of hyperspectral video datasets, we built a 25-band hyperspectral video dataset (the high spectral-spatial-temporal resolution hyperspectral video dataset: the WHU-Hi-H^3 dataset) for target tracking. The experimental results obtained with the WHU-Hi-H^3 dataset confirm the potential of unsupervised deep learning in hyperspectral video target tracking.**

***Index Terms*— Dual-branch network, hyperspectral target tracking, Siamese framework, unsupervised learning.**

I. Introduction

HYPERSPECTRAL imaging technology has evolved over 30 years, and can be used to obtain a data cube with both spatial information and rich spectral information. As a result, hyperspectral imaging technology is now widely used in applications such as geological mapping [1], environmental monitoring [2], vegetation analysis [3], military applications [5], biological applications [6], chemical detection [7], and target detection [8].

A. Brief Description of H^3 Data

The traditional hyperspectral imaging approaches can be divided into whisk broom scanning [9], push broom scanning [10], and "staring"-type scanning [11], and the hyperspectral image data captured by the above sensors can be defined as H^1 (H^1 data: image with only a high spectral resolution) and H^2 data [12]. Specifically, H^1 data, usually obtained by spaceborne and airborne sensors, refers to the hyperspectral image with a high spectral resolution but relative lower spatial resolution, due to the trade-off between spatial and spectral resolution. Recently, with the mature of hyperspectral sensors and platforms, H^2 data, i.e., hyperspectral image data with a high spatial resolution, can be obtained using unmanned aerial vehicle (UAV)-borne hyperspectral systems [13], which further improve the performance of target detection. However, all of the above spectral imaging techniques require spatial scanning or temporal scanning, so a complete spectral cube for a dynamic target cannot be obtained in a single exposure.

With the development of hyperspectral technology, snapshot hyperspectral sensors [14] have now been developed, which have the ability to obtain hyperspectral data cube by real-time imaging. As a result, the snapshot sensors can obtain all the spectral and spatial information of the target in a single exposure. A complete spectral cube of a target can be obtained in one exposure, so that dynamic targets can be captured in high spectral-spatial-temporal resolution (which we refer to as H^3 data). The evolution of data from H^1 data to H^3 data is shown in Fig. 1.

B. Hyperspectral Video Target Tracking

H^3 data provide us with a new data source and the chance to achieve hyperspectral video target tracking. Target tracking [15] is used to predict the size and position of the target in subsequent frames when the target size and position are given in the initial frame of the video sequence. At present, most single-target tracking algorithms are developed based on RGB video data [16]. Video target tracking methods can be

Manuscript received February 14, 2021; revised June 25, 2021 and August 1, 2021; accepted August 22, 2021. Date of publication November 23, 2021; date of current version January 26, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 41771385, Grant 42071350, and Grant 42101327; in part by the Fundamental Research Funds for the Central Universities under Grant 2042021kf0070; and in part by Hubei Provincial Natural Science Foundation of China under Grant 2020CFB275. *(Corresponding author: Yanfei Zhong.)*

Zhenqi Liu, Yanfei Zhong, Meng Shu, and Liangpei Zhang are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: liuzhenqi@whu.edu.cn; zhongyanfei@whu.edu.cn; mengshu@whu.edu.cn; zlp62@whu.edu.cn).

Xinyu Wang is with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: wangxinyu@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2021.3111183

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

simply divided into deep learning-based methods and nondeep learning methods. The most representative nondeep learning video target tracking methods are the methods based on correlation filters, such as the minimum output sum of squared error (MOSSE) filter [17], the kernelized correlation filter (KCF) [18], and the method based on color attributes [19]. The basic idea of correlation filter tracking is to design a filter template which performs correlation operations within the target candidate area, where the position of the maximum output response is the target position of the current frame. These algorithms use features such as the histogram of oriented gradients (HOGs) feature [20] for target tracking. The deep learning-based video target tracking algorithms are represented by the Siamese series of tracking algorithms [21], including SiamFC [22], CFNet [23], the Siamese region proposal network (SiamRPN) [24], and SiamRPN++ [25].

Compared with RGB-based video target tracking, hyperspectral video target tracking is an emerging field. Hyperspectral target tracking is aimed at using the spatial and spectral characteristics of the targets to finely identify targets that are difficult to distinguish. In recent years, the development of snapshot hyperspectral technology has made it possible to obtain hyperspectral videos. At present, the main methods of hyperspectral video target tracking are non deep learning methods. For example, Van Nguyen *et al.* [26] improved the mean-shift algorithm [27], [28] and applied it to hyperspectral video data for target tracking. However, this algorithm only uses the spectral features of the imagery, and the spatial features are not considered. Kandylakis *et al.* [29] used hyperspectral video for multi-target recognition and tracking, but they did not specifically describe how to use the hyperspectral features. Xiong *et al.* [30] proposed a hyperspectral video target tracking algorithm in 2020, which they called material based object tracking in hyperspectral videos (MHT). MHT is based on a background-aware correlation filter (BACF) [31], which uses a 3-D local spectral-spatial histogram of multidimensional gradients (SSHMGs) descriptor (an improved version of the HOG feature, i.e., a 3-D HOG feature) combined with hyperspectral unmixing [32] to obtain mixed features. Qian *et al.* [33] proposed a method of hyperspectral target tracking, which they called convolutional network-based hyperspectral tracking (CNHT), based on the KCF, in which normalized 3-D patches are selected from the target region in the initial frame and fixed convolution kernels are used for the feature extraction in the succeeding frames. The performance of CNHT is, however, limited, due to the fact that it only considers randomly fixed convolution kernels, without training. Meanwhile, the hyperspectral target tracking algorithms based on handcrafted features have a high computational complexity.

To date, the only two publicly available hyperspectral video datasets are those released by Xiong *et al.* [30] and Uzkent *et al.* [34]–[37]. Xiong *et al.* [30] introduced a fully annotated hyperspectral video target tracking dataset in 2020, which contains 35 test video sequences and 40 training video sequences. In total, this dataset contains about 30 000 fully annotated frames. Each video sequence in this dataset [30] is also annotated with the relevant factors out of 11 different attributes [38]. Uzkent *et al.* [34] introduced a synthetic aerial hyperspectral video dataset generated by digital imaging and remote-sensing image generation (DIRSIG) software at 1.42 fps in 2019. However, the video was generated at a very low frame rate from the aerial platform, making it difficult to cover the challenges of target tracking from a ground perspective. At the time of writing, comprehensive hyperspectral target tracking datasets are still lacking.

C. Main Contributions

In this article, a fast unsupervised deep learning-based hyperspectral video target tracking algorithm is proposed (H^3Net), and a new hyperspectral video dataset (the WHU-Hi-H^3 dataset) is introduced. H^3Net is a general framework, which can be used by the Siamese series of tracking algorithms. A spectral branch is added to the template branch and search branch of the Siamese network. In the template branch and search branch, the network is divided into two parts. One part is a spatial branch, and the other is a spectral branch. First, the spatial branch is composed of a two-layer 3×3 convolutional network, which is trained through massive RGB data. The spectral branch extracts spectral features from four layers of 1×1 convolutions (the spectral branch of the template branch contains the spectral attention layer, but the spectral branch of the search branch does not have a spectral attention layer). The trained spatial branches (the spatial branch parameters are frozen during joint training) and spectral branches are then jointly trained to obtain the final model. Finally, the spectral features extracted by the spectral branches and the spatial features extracted by the spatial branches are fused into spectral-spatial features, which are input into a discriminative correlation filter (DCF) [17], [39], [40] for the target tracking. To the best of our knowledge, the framework proposed in this article is the first hyperspectral video target tracking framework based on unsupervised deep learning.

The main contributions of this article are summarized as follows.

1) In this article, the H^3Net framework is proposed for hyperspectral video target tracking. H^3Net is a framework based on a Siamese network. Each branch of the network contains two parts (a spectral branch and a spatial branch), which can calculate the location and size of the target quickly and efficiently. To reduce the amount of manual annotation, the framework uses an unsupervised training method. H^3Net uses samples without any annotation for the training. The bounding box (BBox) is initialized randomly during the training, and then the whole network is trained by the principle of cycle consistency [41].
2) To ensure that the network makes full use of the spatial features of the target, spatial branches are introduced to the Siamese network, which can extract more general spatial features by training the network on a large number of RGB video sequences. On the basis of the spatial branches, spectral branches are added to extract hyperspectral features, and weighted spectral channels are obtained through the attention mechanism of the spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

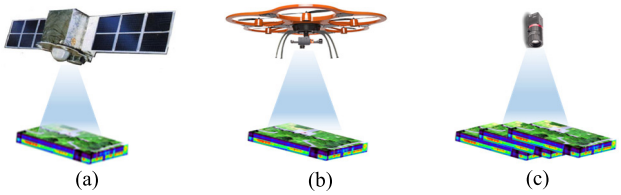


Fig. 1. Evolution of data from H^1 data to H^3 data. (a) H^1 data (hyperspectral data). (b) H^2 data (high spectral-spatial resolution data). (c) H^3 data (high spectral-spatial-temporal resolution data).

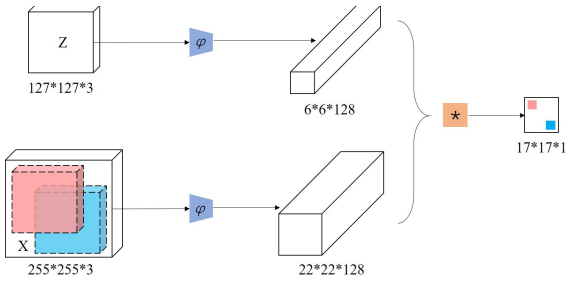


Fig. 2. SiamFC tracker.

channels, which makes the model more accurate. Unlike the spatial branch network, the spectral branch network is trained on a hyperspectral video sequence.

3) A fully annotated hyperspectral video target tracking dataset is presented, which was used to evaluate the hyperspectral video target tracking algorithm proposed in this article.

The rest of this article is organized as follows. Section II briefly introduces the related work in video target tracking. Section III describes the proposed H^{3}Net framework. Section IV describes the WHU-Hi-H^3 dataset. Section V describes the experimental results obtained with the WHU-Hi-H^3 dataset, and further analyzes the proposed modules and the introduced hyperparameters. Finally, Section VI concludes the article.

II. Related Works

A. Unsupervised Learning From Video

Unsupervised learning from video is a new approach which has emerged in recent years. This kind of unsupervised learning, which is also known as self-supervised learning [42], uses training data to train the network, without any annotation. Self-supervised learning from video mainly uses the principle of cycle consistency [41] to train the unannotated training data. Cycle consistency was first proposed in CycleGan [43] to implement image-to-image translation of two domains, without data pairing. In 2019, there were several methods proposed for unsupervised video learning based on the cycle consistency principle. For example, Wang *et al.* [44] trained a deep learning network by forward tracking and backward tracking, and applied the learned model to target segmentation, tracking, and other tasks. Wang *et al.* [45] trained a network for target tracking in a similar way. Both methods expect the initial tracking BBox to be consistent with the tracking BBox from the last frame to the first frame. Dwibedi *et al.* [46] proposed temporal cycle consistency (TCC) learning, in which two frames of two similar videos are aligned, and these videos are then used for the training, without any annotation.

B. Target Tracking Networks Based on Siamese Tracking

The Siamese INstance search Tracker (SINT) [21] was the first target tracking algorithm to be based on a Siamese network. This was also the first time that the target tracking problem had been transformed into a patch block matching problem and realized by a neural network. However, it was the SiamFC tracker [22], as shown in Fig. 2, that really made video target tracking based on a Siamese network popular. The framework of SiamFC is simple and elegant, and its performance reached the state of the art at the time of its release in 2016. In 2017, Valmadre *et al.* [23] proposed CFNet, which adds correlation filtering as a layer to the convolutional neural network (CNN), and derived a forward-propagation formula and back-propagation formula for the correlation filter layer. Wang *et al.* [47] proposed DCFNet, which adds a DCF as a layer into the Siamese network, and they also derived the back-propagation formula for the DCF layer. DCFNet uses a shallow network to extract the features. In 2018, Li *et al.* [24] proposed the SiamRPN [24] method, where, for the first time, an anchor-based method was introduced into target tracking, which greatly improved the processing speed of the deep learning-based target tracking. In 2019, Li *et al.* [25] and Zhang and Peng [48] respectively proposed the SiamRPN++ [25] and SiamDW [48] methods, which solved the problem of the deep learning target tracking algorithms being unable to use a deep network to extract the features for target tracking. Target tracking algorithms based on a Siamese network have been very successful as they can achieve a balance between speed and precision. Siamese network-based target tracking algorithms can also be calculated in real time.

III. H^{3}Net: A Spectral-Spatial-Temporal Resolution Target Tracking Algorithm

In this article, the H^{3}Net target tracking algorithm is proposed to improve the computational efficiency and make use of the hyperspectral features. To increase the amount of available hyperspectral video datasets, we also present the WHU-Hi-H^3 dataset. The main purpose of the H^{3}Net framework is to propose a framework which can allow the existing Siamese target tracking networks based on RGB tracking to be able to use hyperspectral features for the target tracking. As shown in Fig. 3, there are two important parts in the H^{3}Net framework: 1) the spatial branches; and 2) the spectral branches. RGB datasets are used to train the spatial branches. The purpose of this is to be able to use massive RGB datasets to train the model and obtain a general spatial data distribution. Hyperspectral datasets are used to train the spectral branches. The spectral branches can distinguish targets using spectral features. However, due to the lack of hyperspectral video data training samples, and when only

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

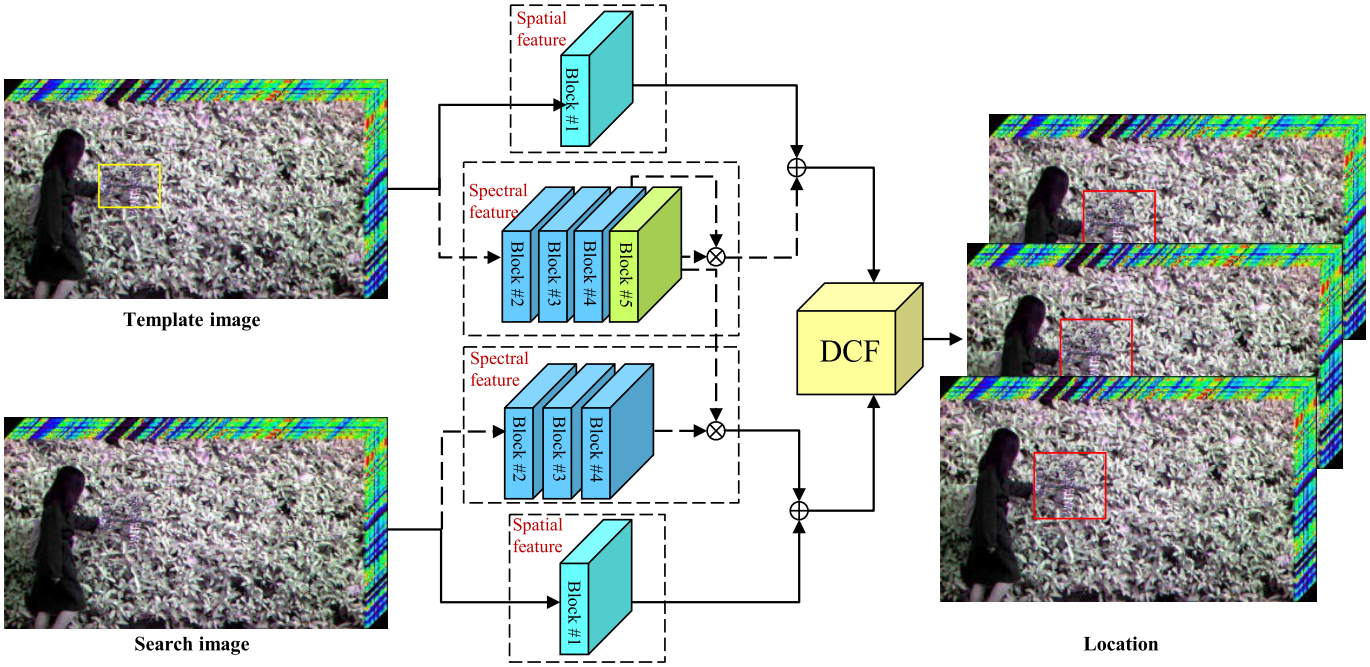


Fig. 3. H^{3}Net: A spectral-spatial-temporal resolution target tracking algorithm.

using the limited hyperspectral video data to train the model, the performance of the model can be very poor. To the best of our knowledge, this is the first time that unsupervised deep learning has been used in hyperspectral target tracking. In deep learning, the supervised deep learning algorithms need a lot of manually annotated data, and the deep learning algorithms for RGB target tracking need about 1 million frames of training data. To solve the problem of the time-consuming and labor-intensive manual tagging, the proposed method uses an unsupervised approach to train the network in an end-to-end manner. To learn more robust spatial features, RGB video datasets are used to train the spatial branches. To utilize hyperspectral features, the spectral branches are trained with hyperspectral video datasets. The whole framework follows a Siamese network structure, and can be used for high-speed target tracking.

A. Unsupervised Training

In recent years, unsupervised deep learning using video datasets has become a hot topic. In the following, we introduce the unsupervised hyperspectral video target tracking training method proposed in this article. As shown in Fig. 4, we use the "cycle consistency" principle to train the model in an unsupervised manner. We first use the principle of cycle consistency to train the spatial branches (for which RGB data are used as the training samples). After the spatial branch training is completed, we use the principle of "cycle consistency" to train the spectral branches (for which hyperspectral data are used as the training samples). When training the spectral branches, the trained spatial branch model is loaded, the parameters of the spatial branches are frozen, and only the gradient of the spectral branches is updated.

1) Consistency Assessment: The first frame gives the BBox (label), which tracks from the first frame to the third frame, which we call forward tracking. It then tracks from the third frame to the first frame, which we call backward tracking. Theoretically, the BBox (with or without target) tracked from the third frame to the first frame in backward tracking should be consistent with the BBox (label) given from the first frame. This is the basis of unsupervised deep learning using video datasets. The reason for using three images for tracking is that, if two images are tracked regularly and consistently, the BBox may appear at the initial position randomly when backward tracking, so that the model cannot learn useful information. The three frames are then used for the cycle consistency training, where the target is tracked directly from the third frame to the first frame (without passing through the second frame). The large displacement difference then means that it is more difficult for the BBox to appear in the initial position randomly, so that the model can learn more useful information. Starting from the initial frame, two frames are randomly selected within ten frames from the initial frame to form a three-frame image group for the cycle consistency training. In this way, through the large displacement difference, the model can learn useful feature information more efficiently. If there are no moving targets in the BBox, this group of images will make little contribution to the loss function of the unsupervised training, so we use a weight function to represent this process

$$M^i_{\text{motion}} = \left\| A^i_{S1} - Y^i_T \right\|_2^2 + \left\| A^i_{S2} - Y^i_{S1} \right\|_2^2 \tag{1}$$

where A^i_{s1} and A^i_{S2} represent the outputs of the second frame and third frame, respectively; and Y^i_T and Y^i_{s1} represent the corresponding ideal Gaussian responses, respectively. The final

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

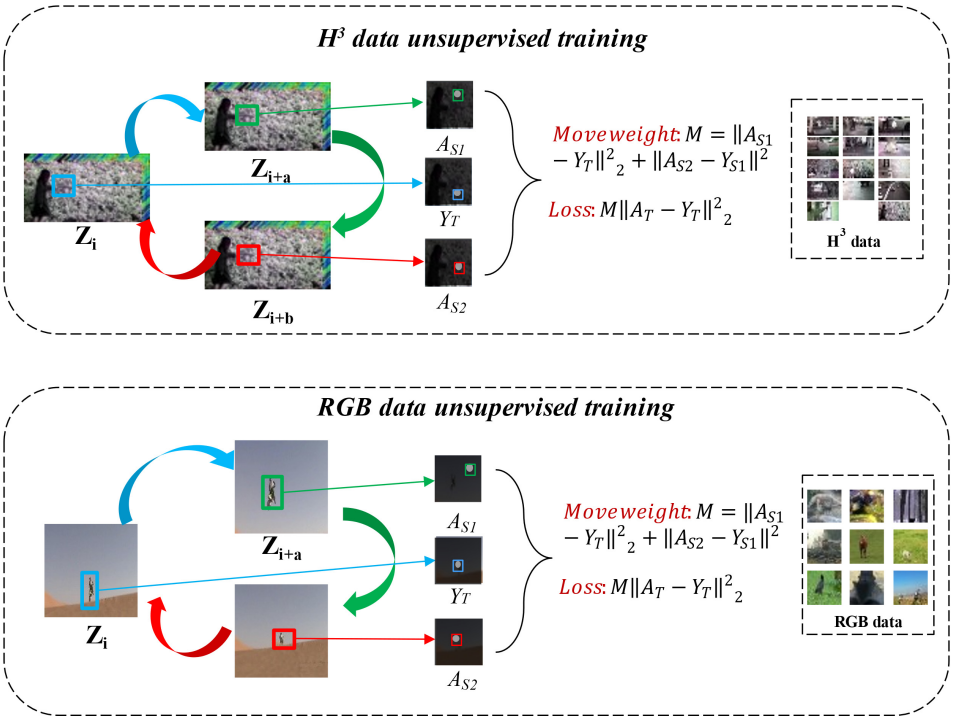


Fig. 4. Unsupervised training.

unsupervised deep learning loss function is

$$L = \frac{1}{n}\sum_{i=1}^{n} M^{i}_{\text{motion}} \left\| A^{i}_{T} - Y^{i}_{T} \right\|^{2}_{2} \tag{2}$$

where n is the number of image pairs in a mini batch, and A^{i}_{T} is the output from the third frame to the first frame. Firstly, we use the cycle consistency principle to train the RGB data in an unsupervised manner, and we then use the cycle consistency principle to train the H^3 data. The unsupervised training method is shown in Fig. 4.

2) Random Initialization of the BBox: In the proposed method, differing from the approach used by Wang *et al.* [45], a randomly initialized BBox is used to track the target. Firstly, as shown in Fig. 5, 200 × 200 pixel image patches are randomly cropped from the original image, and then the image patches are divided into 9 × 9 grids. The search area of the image patch is a 3 × 3 grid, where the search starts from left to right and from top to bottom. A total of 49 moves can traverse the entire image patch. By subtracting the second frame from the first frame of the sequence, the regions (five regions of size 90 × 90 pixels) with the largest difference are obtained, which indicates that the regions may contain dynamic objects. The BBox is then generated by randomly selecting one of the five regions with the largest difference. Finally, the region of 90 × 90 pixels is resized to 125 × 125 pixels, and the image of 125 × 125 pixels is used as the input of the network to train H^3Net.

3) Data Enhancement: Unlike RGB video data, hyperspectral video data cannot be easily obtained from the Internet. It is also difficult to collect hundreds of thousands of frames of training data covering a variety of different scenes in a short period of time. The hyperspectral datasets that can be downloaded online are general hyperspectral remote-sensing image datasets. A single-frame image does not have temporal continuity over a short period of time, so it cannot be used to train a hyperspectral target tracking model based on deep learning. We can therefore use affine transformation, such as image translation, rotation, scale transformation, and other methods, to enhance the imagery, so that the objects in a single hyperspectral video frame can "move." The single video frame can then be used to generate several consecutive video frames containing "moving objects." In this way, a large number of hyperspectral video frames with "moving objects" can be generated. The hyperspectral video training datasets generated in this way can provide more robust information for hyperspectral target tracking based on deep learning.

B. Spatial Branches

The spatial branches are trained on the ILSVRC2015 VID dataset [49]. When the model is tested, three bands from the hyperspectral image are extracted to form a pseudo-color image as the input of the spectral branches. Due to the limited hyperspectral training samples, if there was no spatial branch and only a spectral branch, the performance of the model would be very poor. Massive RGB training samples are used to train the spatial branches, which can enable the model to obtain more general spatial features and make the model more robust. The spatial branches are made up of two convolutional layers, as shown in Fig. 6. The size of the convolution kernel is 3 × 3, and there is an activation function between the two convolutions, which is a rectified linear unit (ReLU) activation function. The last layer is a local response normalization (LRN) layer. When combined with the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

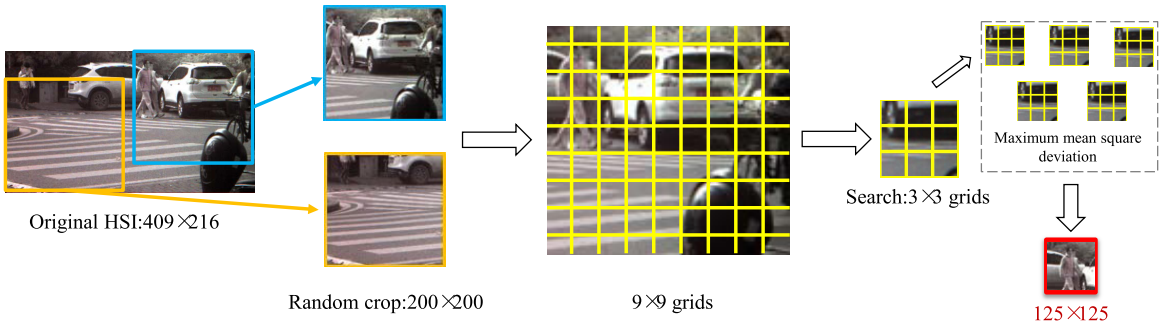


Fig. 5. Random initialization of the BBox.

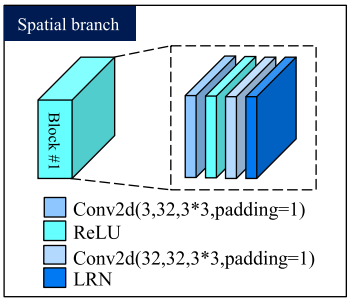


Fig. 6. Network structure for extracting spatial features.

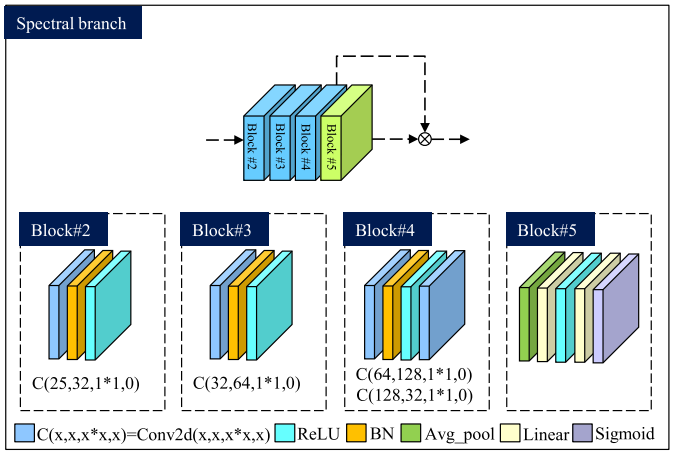


Fig. 7. Network structure for extracting spectral features.

deep learning method, an end-to-end design is realized, and the trained deep learning features replace the handcrafted features as the input of the DCF. Wang *et al.* [47] showed that the DCF can be used as a layer of the deep learning network structure, to deduce the back-propagation formula. Through the shallow network, the target can be accurately located. At the same time, the inference is extremely fast. If only the spatial branches are used for target tracking, a tracking speed of 80 fps can be achieved on the GPU.

C. Spectral Branches

1) Spectral Feature Extraction: The spatial branches are responsible for accurate target location, while the spectral branches makes the network more discriminative. The input of the spectral branches is all the bands of the hyperspectral image. Spectral features are obtained to distinguish targets with similar appearance and different spectra. The spectral branches are made up of four blocks, as shown in Fig. 7.

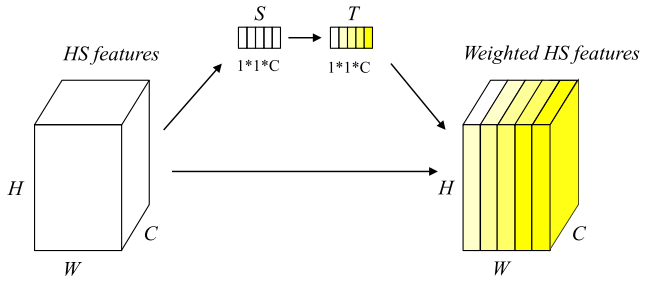


Fig. 8. Spectral attention mechanism.

In Block #2, Block #3, and Block #4, the network structure is almost the same, i.e., a convolutional layer, a batch normalization (BN) layer, and a ReLU activation layer, but in Block #4, a convolutional layer is connected after the activation layer. Using a 1×1 convolution, only the spectral dimension features are learned, so that the spatial position of the image is not damaged, and the network can accurately locate the target. There are four blocks in each spectral branch, which can obtain more discriminative semantic information, without destroying the spatial distribution of the image.

2) Spectral Attention: Block #5 is similar to the SENet architecture [50], and is shown in Fig. 8. A spectral attention mechanism is used to calculate the weight of the spectral channel. For the target tracking, the spectral attention mechanism gives a higher weight to the channels with more influence. The hyperspectral features are the spectral features obtained by Block #2, Block #3, and Block #4. The hyperspectral features form a tensor $HS_features \in R^{C*H*W}$. Firstly, the global description features $S \in R^{C*1*1}$ of the hyperspectral features are calculated by global average pooling. We then use two fully connected layers (with a ReLU activation layer in between the two fully connected layers) and a sigmoid layer to calculate the weight vector $T \in R^{C*1*1}$ of the feature channels. Finally, the original hyperspectral features are multiplied by T to obtain the weighted hyperspectral features. The spectral attention mechanism is only calculated in the template frame, and the weight of the same target in the same scene only needs to be calculated once.

The features obtained from the two branches are both $125 \times 125 \times 32$ in size, so the features can be fused directly, without any upsampling or downsampling. The features of the two branches are fused by simple concatenation. The features from the two branches are fused as inputs of the DCF tracker. In this way, H^{3}Net can obtain more effective features through end-to-end training.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

D. Discriminative Correlation Filter-Based Target Tracking

1) Discriminative Correlation Filtering: DCF-based trackers are the most representative algorithms among the non-deep learning target tracking algorithms. The well-known DCF algorithms are MOSSE [17], KCF [18], the "exploiting the circulant structure of tracking-by-detection with kernels" (CSK) method [51], and the method based on color attributes [19]. In 2017, Wang *et al.* [47] used the DCF as a layer in the deep learning network and derived the back-propagation formula for the DCF layer. The DCF layer and deep learning feature extraction network were then cascaded together for end-to-end training. In the proposed approach, the DCF layer is used to calculate the position and size of the target in the subsequent frames.

In the DCF, the loss function is

$$\min \left\| \sum_{l=1}^{D} w^l * \varphi^l(x) - h \right\|^2 + \lambda \sum_{l=1}^{D} \left\| w^l \right\|^2 \tag{3}$$

where w^l is the filter, l is the number of channels of the filter, $\varphi^l(x) \in R^{M*N*D}$ are the features obtained through the spectral branches and spatial branches, and $h \in R^{M*N}$ is the ideal output.

w^l can be calculated by

$$\hat{w}^l = \frac{\hat{\varphi}^l(x) \odot \hat{h}^*}{\sum_{k=1}^{D} \hat{\varphi}^k(x) \odot (\hat{\varphi}^k(x))^* + \lambda} \tag{4}$$

where $\hat{w}$ is the Fourier transform of w, h^* is the conjugate value of h, and $\odot$ represents the dot product.

The features of the new frame are obtained by the neural network, and the response value R is calculated by the feature. The position of the maximum response value is the position of the target in the new frame

$$R = F^{-1}\left(\sum_{l=1}^{D} \left(\hat{w}^l \right)^* \odot \hat{\varphi}^l(z) \right). \tag{5}$$

2) Online Model Update: During online tracking, the tracking target changes with time, i.e., the size, illumination, location, etc., change during the tracking. Therefore, the filter w needs to be updated during the tracking to adapt to the changes of the target. The optimization problem in (3) can be formulated in an incremental mode as

$$\sum_{t=1}^{p} \alpha_t \left(\left\| \sum_{l=1}^{D} w_p^{l\,*} \varphi^l(x_t) - h \right\|^2 + \lambda \sum_{l=1}^{D} \left\| w_p^l \right\|^2 \right) \tag{6}$$

where parameter $\alpha_t \geq 0$.

Equation (4) can be extended to a time series

$$\hat{w}_p^l = \frac{\sum_{t=1}^{p} \alpha_t \hat{\varphi}^l(x_t) \odot \hat{h}^*}{\sum_{t=1}^{p} \alpha_t \left(\sum_{k=1}^{D} \hat{\varphi}^k(x_t) \odot (\hat{\varphi}^k(x_t))^* + \lambda \right)}. \tag{7}$$

IV. WHU-Hi-H3 Dataset

We referred to the existing RGB video datasets, including OTB100 [38], LaSOT [52], and Need for Speed (NFS) [53], to construct the WHU-Hi-H^3 dataset (a high spectral-spatial-temporal resolution dataset). We also referred to a large-scale aerial vehicle classification dataset [34] and a ground-based hyperspectral dataset [30]. As shown in Fig. 9 and Table I, the WHU-Hi-H^3 dataset contains 6969 frames of annotated hyperspectral video data. The spatial size is 409 × 216, and the band range is from 600 to 900 nm, with a total of 25 bands. To prove that hyperspectral video data can provide more useful features than RGB video data in some specific scenes, we designed a number of camouflaged target scenes. In the WHU-Hi-H^3 dataset, we designed the following nine scenes: cars, coin, bracelet, bag, pedestrians, persons, park, green box, and shadow. These scenes are respectively composed of 1746 frames, 342 frames, 500 frames, 500 frames, 1419 frames, 676 frames, 451 frames, 62 frames, and 1273 frames. The targets to be tracked are, respectively, vehicles, coin, bracelet, school bag, humans, humans, humans, green box, and humans. These nine scenes are divided into 69 sequences. Most of the objects are similar to their surroundings, i.e., they are camouflaged targets. The scenes with camouflaged targets can be used to verify whether an algorithm works better with hyperspectral video data than with RGB video data. We also designed other challenging scenarios for target tracking, including: scenes with illumination variation (IV); scenes with non-rigid object deformation (DEF); scenes where the target is partially or fully occluded (OCC); scenes with scale variation (SV); scenes with motion blur (MB), where the target region is blurred due to the motion of the target or camera; scenes where some portion of the target is out of view (OV); scenes with background clutter (BC), where the background near the target has a similar color or texture to the target; scenes with low resolution (LR), where the number of pixels inside the ground-truth BBox is less than 400; and scenes with fast motion (FM). The IV, DEF, SV, BC, LR, FM, and other challenges degrade the performance of RGB algorithms, but the spectral information of hyperspectral video data does not change in these challenges, and the hyperspectral video data can provide additional spectral information about the target.

V. Experimental Results and Analysis

A. Experimental Settings

1) Experimental Details: The training datasets and the test datasets were made up of spectral video data captured by a 25- channel imec snapshot spectral camera. The spatial size was 409 × 216. The frame rate of the hyperspectral video data was 24 fps and the spectral range was 600–1000 nm. The hyperspectral training dataset was made up of 22 753 frames of hyperspectral video data. The RGB dataset was the ILSVRC2015 VID dataset [49]. The unsupervised method described in Section III was used to train H^{3}Net. We trained the models for 50 epochs on an NVIDIA TITAN X GPU with a stochastic gradient descent (SGD) optimizer. The batch size was set to 32, and the learning rate was set to 0.01. After the training, only forward tracking was needed to obtain the tracking results. In the testing phase, the input of the entire models was only hyperspectral images. The input of the spatial branches of the models was a pseudo-color image composed of three bands from the hyperspectral image (band numbers [1, 16, 25]). The input of the spectral branches of the models

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.



Fig. 9. WHU-Hi-H^3 dataset.

was the entire hyperspectral image, including all bands. The experiments were run in Python (PyTorch) on an r720 Lenovo notebook with a Core i7 CPU and an NVIDIA GeForce GTX 1050 GPU. The processing speed of the program was 6.6 fps.

2) Metrics: The evaluation standard for the hyperspectral target tracking dataset (WHU-Hi-H3) refers to the existing evaluation standard for RGB video datasets, which is known as the Object Tracking Benchmark (OTB [38]), i.e., the average pixel error (APE) and the average overlap rate (AOR). The APE, as the name implies, is the error value based on the pixel distance between the predicted target center position and the center position of the BBox of the ground truth. The larger the value, the greater the error. The AOR is the AOR between the BBox obtained by the algorithm in each frame of the video and the BBox of the ground truth. The larger the value, the better the effect. The APE and the AOR were respectively used to generate precision plots and success plots. We also use the DP20pixels metric, which is the percentage of video frames whose distance between the center point of the BBox predicted by the tracking algorithm and the center point of the ground truth is less than 20 pixels. In Fig. 10, in the precision plots, if we take the value of 20 for the horizontal axis and the value of 0.5 for the vertical axis as examples, it can be seen that frames within the distance between the center of the BBox of the tracking result and the center of the BBox of the ground truth of less than or equal to 20 pixels account for 50% of all the frames. In the success plots, taking the value of 0.5 for the horizontal axis and the value of 0.5 for the vertical axis as examples, it can be seen that the frames for which the overlap rate between the BBox of the tracking result and the BBox of the ground truth is more than 50% account for 50% of all the frames.

B. Experiment 1: Hyperspectral Video Target Tracking Experiment

In this experiment, we compared all the current hyperspectral video target tracking methods, i.e., CNHT [33], DeepHKCF [34], and the proposed H^{3}Net method. As a target tracking algorithm based on RGB video, the unsupervised deep tracking (UDT) method proposed in 2019 was the first end-to-end trained target tracking algorithm based on unsupervised deep learning [45]. UDT_25bands represents the use of the UDT framework to train the hyperspectral video data, but the number of input bands of UDT is changed from the three bands of RGB video data to the 25 bands of hyperspectral video data. The CNHT algorithm simply uses convolution to extract features, and the normalized 3-D patches selected from the target region in the initial frame and random fixed convolution kernels are generated for feature extraction in the succeeding frames. DeepHKCF uses pseudo-color video to extract features, and is an algorithm for training on RGB video that does not use real hyperspectral features.

Fig. 10 shows the performance of each algorithm. The results show that CNHT achieves an inferior accuracy, due to the fact that it only considers fixed positive samples when learning the convolutional filters. Without negative patches in the surrounding background, the features produced by the randomly fixed convolutional filters without training are not discriminative enough to learn a robust model, which significantly worsens the predicted object location. DeepHKCF

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

TABLE I
BASIC PROPERTIES OF THE WHU-HI-H3 DATASET

Scenes	Spectral resolution	Attributes	Spatial size	Annotated image number
9	25 bands, 600–1000 nm	BC, LR, IV, SV, OCC, …	409 × 216	6969

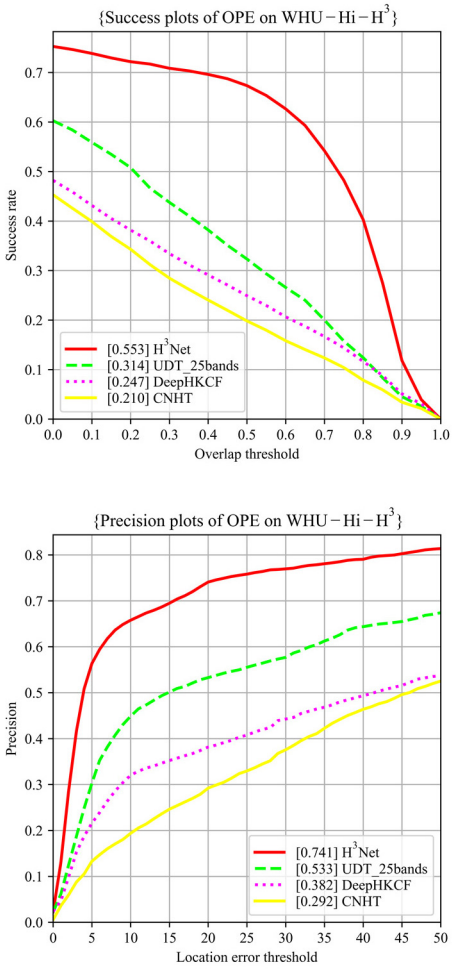


Fig. 10. Test results of the hyperspectral algorithms with the WHU-Hi-H^3 dataset.

uses VGGNet to extract features. As a non-end-to-end training framework, DeepHKCF is pre-trained on RGB video data, and is not trained on hyperspectral video data. DeepHKCF uses a pre-trained model to obtain a better performance than CNHT, without training. UDT_25bands uses hyperspectral video for end-to-end training, so that the accuracy is higher than that of DeepHKCF, which uses RGB data for pre-training. The spatial branches of H^{3}Net are trained with RGB video data, so that spatial features with strong robustness are obtained. Meanwhile, the spectral branches of H^{3}Net are trained by hyperspectral video to obtain the spectral features. In H^{3}Net, the spatial features are fused with the spectral features, and the fused features are then used as the input of the DCF to predict the location and size of the targets. As a result, H^{3}Net shows a much better performance than UDT_25bands, which only uses hyperspectral video for the training. H^{3}Net solves the problem of the limited hyperspectral video training set samples, to a certain extent. At the same time, H^{3}Net uses a shallow network to extract the features, and obtains the best

TABLE II
TEST RESULTS FOR THE HYPERSPECTRAL ALGORITHMS ON THE WHU-HI-H^3 DATASET

WHU-Hi-H^3 dataset			
Tracker	AUC	DP20pixels	FPS
CNHT [33]	0.210	0.292	0.49
DeepHKCF [34]	0.247	0.382	1.1
UDT_25bands	0.314	0.533	**6.8**
H^{3}Net	**0.553**	**0.741**	6.6

TABLE III
PARAMETERS AND FLOPS OF H^{3}NET

H^{3}Net		
Branch	FLOPS	Parameters
Spatial branch	158.5M	0.01M
Spectral branch	248M	0.02M

tracking performance and the second-fastest processing speed. As shown in Table II, the speed of CNHT is the slowest, the speed of DeepHKCF is at a medium level, the speed of UDT_25bands is the fastest, and H^{3}Net ranks second. In terms of accuracy, H^{3}Net significantly outperforms the other hyperspectral algorithms. At the same time, in terms of processing speed, H^{3}Net is only slightly inferior to UDT_25bands, but is significantly ahead of the other algorithms. The parameters and floating-point operations per second (FLOPS) of the model are listed in Table III, where it can be seen that the FLOPS and parameters of the two branches of H^{3}Net are very small.

Overall, from the visualization results shown in Figs. 11 and 12, H^{3}Net shows the best performance among the different hyperspectral target tracking algorithms.

C. Experiment 2: Effectiveness of the RGB and Spectral Characteristics

We tested the H^{3}Net_3bands algorithm on pseudo-color video and the H^{3}Net algorithm on hyperspectral video. The pseudo-color video was synthesized by extracting three bands from the hyperspectral video data, to simulate RGB data. The spectral branches of H^{3}Net_3bands were trained with a pseudo-color training set, and the spectral branches of H^{3}Net were trained with a hyperspectral dataset. There was no change in the network structure and parameters of the H^{3}Net algorithm, except for the numbers of input bands being different.

It can be seen from the results shown in Fig. 13 that, in the case of the same amount of training data, the results obtained using hyperspectral training are better than those

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.



Fig. 11. Tracking results in the scene of persons wearing green clothes.

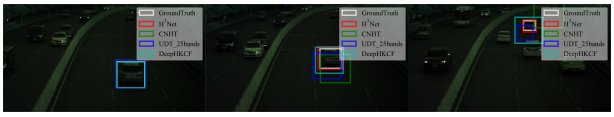


Fig. 12. Tracking results in the vehicle scene.

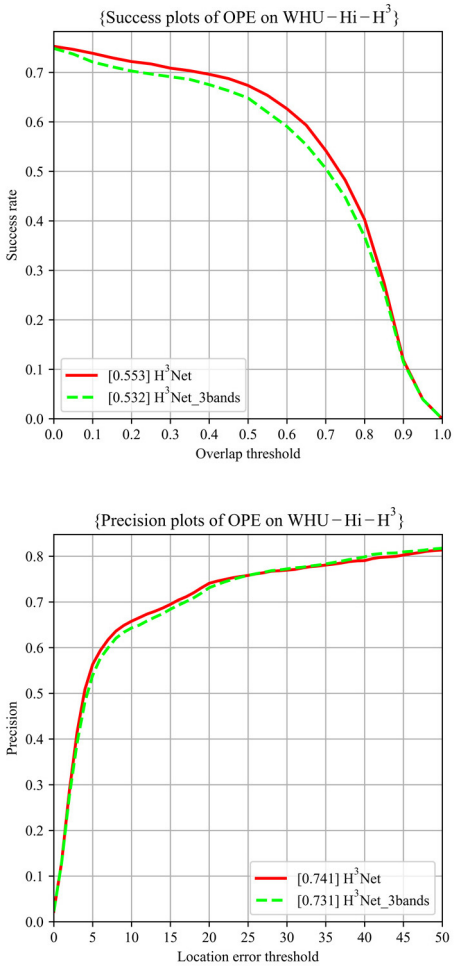


Fig. 13. Test results of the H^{3}Net and H^{3}Net_3bands algorithms with the WHU-Hi-H^3 dataset.

obtained using pseudo-color training, which proves that the hyperspectral data features are more effective than the RGB data features.

D. Experiment 3: Effectiveness Evaluation for the Data Enhancement Method

With regard to the training methods for RGB-based target tracking, the RGB-based deep learning target tracking algorithms need millions of frames of video data as training samples. Compared with hyperspectral video data, RGB video data are much easier to obtain. A large number of RGB videos can be downloaded online. However, hyperspectral video can only be acquired by snapshot spectral cameras. It is therefore impossible to obtain a wide range of hyperspectral video data.

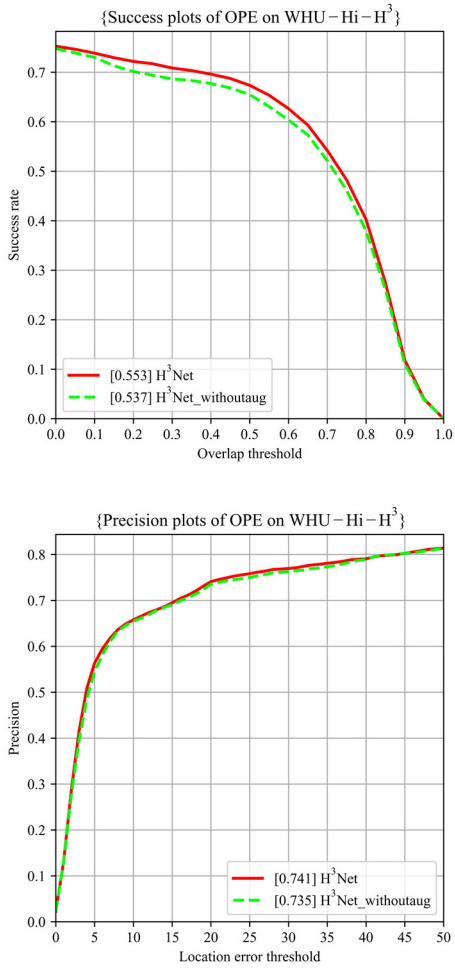


Fig. 14. Data enhancement experiment comparison.

As a result, hyperspectral video training samples for training hyperspectral target tracking algorithms are extremely scarce.

We increased the training samples of the WHU-Hi-H3 dataset to 68 131 frames using the data enhancement method described in Section III. We then used these training samples to train H^{3}Net in an end-to-end manner. We also used the samples without data enhancement to train H^{3}Net. In Fig. 14, H^{3}Net represents the model trained with data augmentation, and H^{3}Net_withoutaug represents the model trained without data augmentation. In the precision plots, it can be seen that the DP20pixels value increases from 73.5% to 74.1% after using data enhancement. In the success plots, the area under the curve (AUC) score of the algorithm trained with data enhancement is improved by 1.6%. It can therefore be concluded that the data enhancement method proposed in this article is effective for hyperspectral video target tracking algorithms based on deep learning. The data enhancement also provides a solution to the shortage of training samples for hyperspectral video data.

E. Experiment 4: Effectiveness Evaluation of the Dual-Branch Siamese Network

Because of the huge volume of RGB video data, a dual-branch Siamese network (H^{3}Net) is proposed. The spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

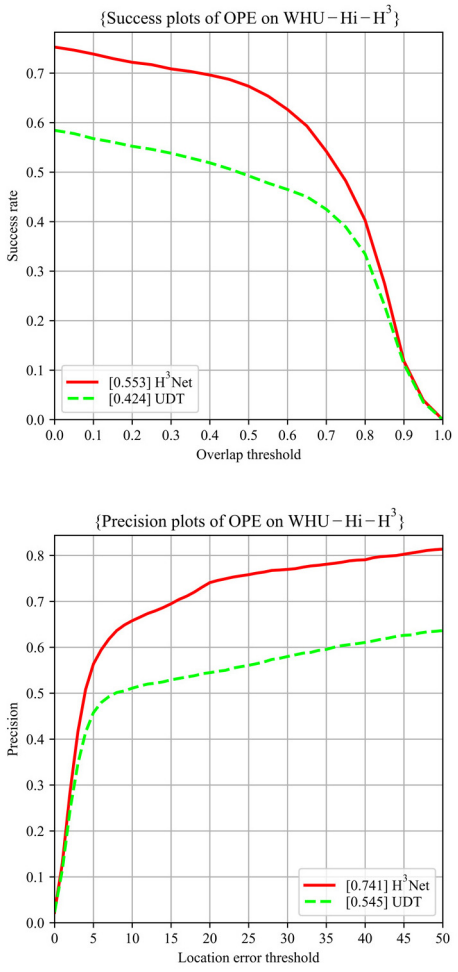


Fig. 15. Effectiveness evaluation of the dual-branch Siamese network.

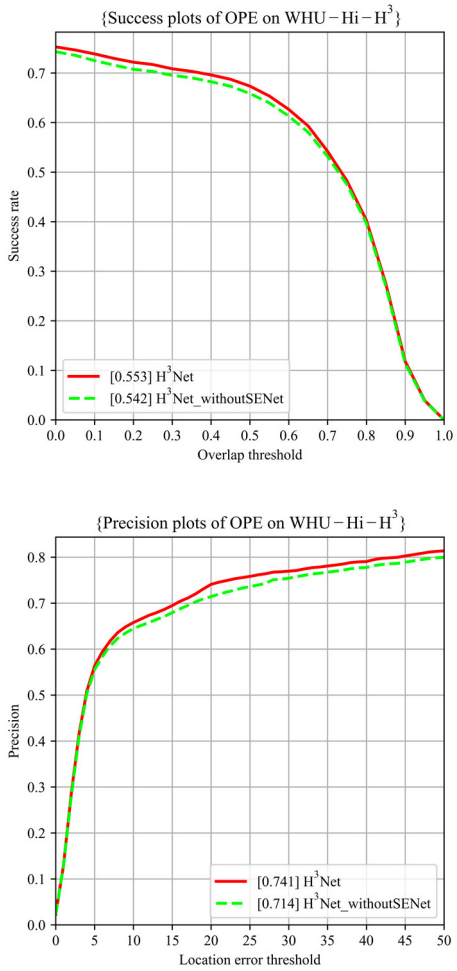


Fig. 16. Spectral attention sensitivity analysis.

branches of the dual-branch Siamese network can train the spatial features using the abundant RGB video datasets, and the spectral branches branch can train the spectral features using hyperspectral video datasets. Finally, the two features are fused to obtain more robust and discriminative features.

The results of H^{3}Net and the UDT method are compared in Fig. 15, where UDT represents the result of using the model trained on RGB video data provided by the authors of the UDT method. In the success plots, the AUC score of H^{3}Net (55.3%) outperforms the AUC score of UDT (42.4%) by 12.9%. In the precision plots, the DP20pixels value is improved from 54.5% to 74.1%, which is an improvement of 19.6%. The test results obtained with the WHU-Hi-H3 dataset show that H^{3}Net trained with RGB video and hyperspectral video shows a better performance than UDT trained with RGB video. Meanwhile, the features learned by H^{3}Net with a randomly initialized BBox are superior in tracking performance to the features learned by UDT with only a centrally initialized BBox.

F. Experiment 5: Spectral Attention Sensitivity Analysis

We also assessed the effectiveness of the spectral attention mechanism. The attention mechanism is an algorithm that was proposed in 2017 by Hu *et al.* [50]. In the task of image classification, the attention mechanism is added to each backbone, so that the classification accuracy is improved accordingly. This experiment was aimed at proving that the spectral attention mechanism can improve the accuracy of a hyperspectral video target tracking algorithm.

From the experimental results shown in Fig. 16, it can be seen that the spectral attention mechanism effectively improves the performance of the tracking algorithm. In the success plots, the AUC score of the algorithm is improved from 54.2% to 55.3%, which is an improvement of 1.1%. In the precision plots, the DP20pixels value of the algorithm is improved from 71.4% to 74.1%, which is an improvement of 2.7%. The spectral attention mechanism improves the performance of the hyperspectral target tracking algorithm with minimal extra computational cost.

G. Experiment 6: Comparison of Fixed Initialization and Random Initialization for the BBox

The UDT algorithm initializes the BBox in the image center, while H^{3}Net initializes the BBox randomly in the image. This experiment was aimed at comparing the two unsupervised BBox initialization methods. We used the 68 161 enhanced frames of hyperspectral video data for the training. H^{3}Net represents the random initialization BBox method, as proposed in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

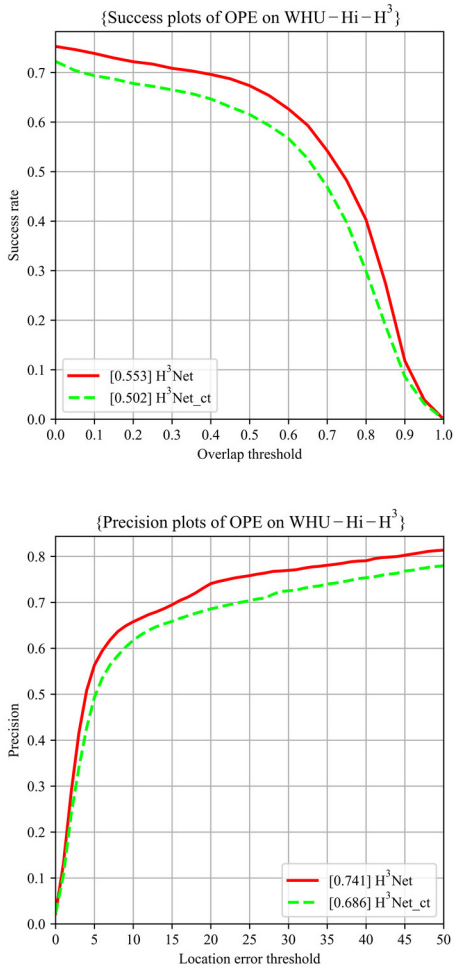


Fig. 17. Results of fixed initialization and random initialization.

this article, while H^{3}Net_ct represents initializing the BBox in the image center.

From the experimental results shown in Fig. 17, it can be seen that the method of random initialization can obtain better tracking results than the method of initializing the BBox in the image center. In the success plots, the AUC score of the algorithm is improved from 50.2% to 55.3%, which is an improvement of 5.1%. In the precision plots, the DP20pixels value of the algorithm is improved from 68.6% to 74.1%, which is an improvement of 5.5%. The experimental results show that the effect of random initialization is better than the effect of the BBox being fixed in the image center. The main reason for this is that a large number of the dynamic targets in the training set are not in the center of the image, so that the method of BBox initialization in the image center learns the characteristics of some static targets, which has a negative impact on the model.

H. Experiment 7: Ablation Study

We verified the impact of each block of the model on the final performance of the algorithm, and we also verified the impact of using different datasets to train the model. The ablation study results are listed in Table IV. When only hyperspectral data are used to train Block #1, the AUC score is only 37.3%. When the ILSVRC2015 VID RGB dataset [49] is used to train Block #1, the AUC score is 51.7%. The reason for this is that there are very few hyperspectral training samples, and the performance of models trained directly using hyperspectral training samples is worse than that of models trained using RGB training samples. Therefore, we propose H^{3}NET, which adds spectral branches on the basis of spatial branches, to achieve a discriminative model.

TABLE IV
ABLATION STUDY

Method	Training data	AUC	Δ AUC
B1	Hyperspectral	37.3%	-
B1	RGB	51.7%	+14.4%
B1+B2	Hyperspectral+RGB	53.0%	+15.7%
B1+B2+B3	Hyperspectral+RGB	53.5%	+16.2%
B1+B2+B3+B4	Hyperspectral+RGB	54.2%	+16.9%
B1+B2+B3+B4+B5	Hyperspectral+RGB	55.3%	+18.0%

B1: Block #1, B2: Block #2, B3: Block #3, B4: Block #4, B5: Block #5

VI. CONCLUSION

In this article, we have proposed an unsupervised deep learning method (H^{3}Net) for target tracking based on hyperspectral video. We have also presented the high spectral-spatial-temporal resolution hyperspectral video dataset (WHU-Hi-H^3) for hyperspectral video target tracking. To reduce the work of manual annotation, an unsupervised deep learning framework is used in H^{3}Net. To enable the model to learn features that are useful for the tracking, a method of randomly initializing the BBox is used in H^{3}Net. Due to the scarcity of hyperspectral video datasets, learning features using hyperspectral data is essentially a small sample learning problem. We therefore proposed a dual-branch network structure based on a Siamese network, which uses RGB video data and hyperspectral video data to learn the spatial and spectral features, respectively, and then the fused features are used for the target tracking. This solves the problem of small sample learning for hyperspectral video target tracking, to a certain extent. A spectral attention mechanism is also used in the spectral branches to further improve the effectiveness of the spectral features during the tracking. The spectral attention mechanism is only used in the first frame, which does not bring additional computational complexity to the algorithm. We also proposed a data enhancement scheme for the hyperspectral video data, which further improves the performance of the algorithm. At the same time, to address the lack of test data for hyperspectral video, we introduced the high spectral-spatial-temporal resolution hyperspectral target tracking dataset (the WHU-Hi-H^3 dataset). In the results of the experiments conducted on the WHU-Hi-H^3 dataset, the H^{3}Net algorithm proposed in this article obtained the best target tracking performance and a fast processing speed.

In the future, we will explore better-performing hyperspectral video target tracking algorithms, and we will build an even larger hyperspectral video dataset. We hope that the algorithm and dataset introduced in this article will make some positive contributions to the field of hyperspectral video target tracking.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] F. A. Kruse, J. W. Boardman, and J. F. Huntington, "Comparison of airborne hyperspectral data and EO-1 Hyperion for mineral mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 6, pp. 1388–1400, Jun. 2003.

[2] E. L. Hestir *et al.*, "Identification of invasive vegetation using hyperspectral remote sensing in the California delta ecosystem," *Remote Sens. Environ.*, vol. 112, pp. 4034–4047, Nov. 2008.

[3] D. Haboudane, J. R. Miller, E. Pattey, P. J. Zarco-Tejada, and I. B. Strachan, "Hyperspectral vegetation indices and novel algorithms for predicting green LAI of crop canopies: Modeling and validation in the context of precision agriculture," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 337–352, 2004.

[4] S. Kacenjar and M. Honvedel, "Atmospheric characterization using space-based hyperspectral imaging systems," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Anchorage, AK, USA, Sep. 2004, pp. 4199–4202.

[5] M. Shimoni, R. Haelterman, and C. Perneel, "Hypersectral imaging for military and security applications: Combining myriad processing and sensing techniques," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 101–117, Jun. 2019.

[6] G. L. Anderson, R. I. Carruthers, S. Ge, and P. Gong, "Cover: Monitoring of invasive tamarix distribution and effects of biological control with airborne hyperspectral remote sensing," *Int. J. Remote Sens.*, vol. 26, no. 12, pp. 2487–2489, Jun. 2005.

[7] S. Boudreau *et al.*, "Chemical detection with hyperspectral lidar using dual frequency combs," *Opt. Exp.*, vol. 21, no. 6, pp. 7411–7418, Mar. 2013.

[8] Q. Zuo, B. Guo, H. Shen, M. Yang, and K. Cheng, "An improved target detection algorithm for camouflaged targets," in *Proc. 36th Chin. Control Conf. (CCC)*, Dalian, China, Jul. 2017, pp. 11478–11482.

[9] Z. Wang and X. Xiong, "Band-to-band misregistration of the images of MODIS onboard calibrators and its impact on calibration," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 4, pp. 2136–2143, Apr. 2017.

[10] S. Ortega *et al.*, "Hyperspectral push-broom microscope development and characterization," *IEEE Access*, vol. 7, pp. 122473–122491, Aug. 2019.

[11] N. Gupta, "Development of staring hyperspectral imagers," in *Proc. IEEE Appl. Imag. Pattern Recognit. Workshop (AIPR)*, Oct. 2011, pp. 1–8.

[12] Y. Zhong *et al.*, "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.

[13] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.

[14] Z. Xiong, L. Wang, H. Li, D. Liu, and F. Wu, "Snapshot hyperspectral light field imaging," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 6873–6881.

[15] A. E. Mitchell, G. E. Smith, K. L. Bell, and M. Rangaswamy, "Single target tracking with distributed cognitive radar," in *Proc. IEEE Radar Conf. (RadarConf)*, Seattle, WA, USA, May 2017, pp. 0285–0288.

[16] H. Huang and S. Lei, "Guided cross-component prediction for RGB video coding," in *Proc. Data Compress. Conf.*, Snowbird, UT, USA, Mar. 2018, pp. 80–86.

[17] D. Bolme, J. R. Beveridge, B. A. Draper, and Y. M. Lui, "Visual object tracking using adaptive correlation filters," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, San Francisco, CA, USA, Jun. 2010, pp. 2544–2550.

[18] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "High-speed tracking with kernelized correlation filters," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 583–596, Mar. 2015.

[19] M. Danelljan, F. S. Khan, M. Felsberg, and J. Van De Weijer, "Adaptive color attributes for real-time visual tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 1090–1097.

[20] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, San Diego, CA, USA, Jun. 2005, pp. 886–893.

[21] R. Tao, E. Gavves, and A. W. M. Smeulders, "Siamese instance search for tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 1420–1429.

[22] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, and P. H. S. Torr, "Fully-convolutional Siamese networks for object tracking," in *Proc. ECCV Workshops*, Amsterdam, Netherlands, 2016, pp. 850–865.

[23] J. Valmadre, L. Bertinetto, J. Henriques, A. Vedaldi, and P. H. S. Torr, "End-to-end representation learning for correlation filter based tracking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 5000–5008.

[24] B. Li, J. Yan, W. Wu, Z. Zhu, and X. Hu, "High performance visual tracking with Siamese region proposal network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 8971–8980.

[25] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, and J. Yan, "SiamRPN++: Evolution of Siamese visual tracking with very deep networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 4277–4286.

[26] H. Van Nguyen, A. Banerjee, and R. Chellappa, "Tracking via object reflectance using a hyperspectral video camera," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (Workshops)*, San Francisco, CA, USA, Jun. 2010, pp. 44–51.

[27] Y. Cheng, "Mean shift, mode seeking, and clustering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 17, no. 8, pp. 790–799, Aug. 1995.

[28] D. Comaniciu, V. Ramesh, and P. Meer, "Real-time tracking of non-rigid objects using mean shift," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Hilton Head Island, SC, USA, Jun. 2000, pp. 142–149.

[29] Z. Kandylakis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Multiple object tracking with background estimation in hyperspectral video sequences," in *Proc. 7th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Tokyo, Japan, Jun. 2015, pp. 1–4.

[30] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Trans. Image Process.*, vol. 29, pp. 3719–3733, Jan. 2020.

[31] H. K. Galoogahi, A. Fagg, and S. Lucey, "Learning background-aware correlation filters for visual tracking," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 1144–1152.

[32] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[33] K. Qian, J. Zhou, F. Xiong, and H. Zhou, "Object tracking in hyperspectral videos with convolutional features and kernelized correlation filter," in *Proc. Int. Conf. Smart Multimedia*, Toulon, France, 2018, pp. 308–319.

[34] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Tracking in aerial hyperspectral videos using deep kernelized correlation filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 449–461, Jan. 2019.

[35] B. Uzkent, A. Rangnekar, and M. J. Hoffman, "Aerial vehicle tracking by adaptive fusion of hyperspectral likelihood maps," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Honolulu, HI, USA, Jul. 2017, pp. 233–242.

[36] B. Uzkent, M. J. Hoffman, and A. Vodacek, "Spectral validation of measurements in a vehicle tracking DDDAS," *Procedia Comput. Sci.*, vol. 51, pp. 2493–2502, Jan. 2015.

[37] B. Uzkent, M. J. Hoffman, A. Vodacek, J. P. Kerekes, and B. Chen, "Feature matching and adaptive prediction models in an object tracking DDDAS," *Procedia Comput. Sci.*, vol. 18, pp. 1939–1948, Jan. 2013.

[38] Y. Wu, J. Lim, and M. H. Yang, "Object tracking benchmark," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1834–1848, Sep. 2015.

[39] M. Danelljan, G. Häger, F. S. Khan, and M. Felsberg, "Discriminative scale space tracking," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 8, pp. 1561–1575, Aug. 2017.

[40] Y. Li and J. Zhu, "A scale adaptive kernel correlation filter tracker with feature integration," in *Proc. ECCV Workshops*, Zürich, Switzerland, 2014, pp. 254–265.

[41] L. Wu, Y. Wang, and L. Shao, "Cycle-consistent deep generative hashing for cross-modal retrieval," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1602–1612, Apr. 2019.

[42] X. Wang and A. Gupta, "Unsupervised learning of visual representations using videos," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Santiago, Chile, Dec. 2015, pp. 2794–2802.

[43] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 2242–2251.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.

[44] X. Wang, A. Jabri, and A. A. Efros, "Learning correspondence from the cycle-consistency of time," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 2561–2571.

[45] N. Wang, Y. Song, C. Ma, W. Zhou, W. Liu, and H. Li, "Unsupervised deep tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 1308–1317.

[46] D. Dwibedi, Y. Aytar, J. Tompson, P. Sermanet, and A. Zisserman, "Temporal cycle-consistency learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 1801–1810.

[47] Q. Wang, J. Gao, J. Xing, M. Zhang, and W. Hu, "DCFNet: Discriminant correlation filters network for visual tracking," 2017, *arXiv:1704.04057*. [Online]. Available: http://arxiv.org/abs/1704.04057

[48] Z. Zhang and H. Peng, "Deeper and wider Siamese networks for real-time visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 4586–4595.

[49] O. Russakovsky *et al.*, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.

[50] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.

[51] J. F. Henriques *et al.*, "Exploiting the circulant structure of tracking-by-detection with kernels," in *Proc. ECCV Workshops*, Firenze, Italy, 2012, pp. 702–715.

[52] H. Fan *et al.*, "LaSOT: A high-quality benchmark for large-scale single object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 5369–5378.

[53] H. K. Galoogahi, A. Fagg, C. Huang, D. Ramanan, and S. Lucey, "Need for speed: A benchmark for higher frame rate object tracking," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Venice, Italy, Oct. 2017, pp. 1134–1143.



Zhenqi Liu received the B.S. and M.S. degrees from North University of China, Taiyuan, China, in 2012 and 2018, respectively. He is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

His research interests include computer vision and hyperspectral object tracking.



Yanfei Zhong (Senior Member, IEEE) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) research group, Wuhan University. He has published more than 100 research articles in international journals, such as *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is currently a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He won the Second-Place Prize in 2013 IEEE GRSS Data Fusion Contest and the Single-View Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is also serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and the *International Journal of Remote Sensing*.



Xinyu Wang received the B.S. degree in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2014, and the Ph.D. degree in communication and information systems from the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University, in 2019.

Since 2019, he has been an Associate Research Fellow with the School of Remote Sensing and Information Engineering, Wuhan University. His research interests include hyperspectral unmixing, detection, restoration, and tracking.



Meng Shu received the B.S. degree from the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China, in 2019. She is currently pursuing the master's degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University.

Her research interests include satellite video object detection and satellite video object segmentation.



Liangpei Zhang (Fellow, IEEE) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He is currently the "Chang-Jiang Scholar" Chair Professor appointed by the Ministry of Education of China in the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He was a Principal Scientist of the China State Key Basic Research Project (2011–2016) appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. He has published more than 700 research articles and five books. He is also the Institute for Scientific Information (ISI) highly cited author. He is the holder of 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is also a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams won the top three prizes of the IEEE Geoscience and Remote Sensing Society (GRSS) 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) Student Paper Contest in recent years. He is also the Founding Chair of the IEEE GRSS Wuhan Chapter. He also serves as an associate editor or an editor for more than ten international journals. He is also serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 12:22:04 UTC from IEEE Xplore. Restrictions apply.